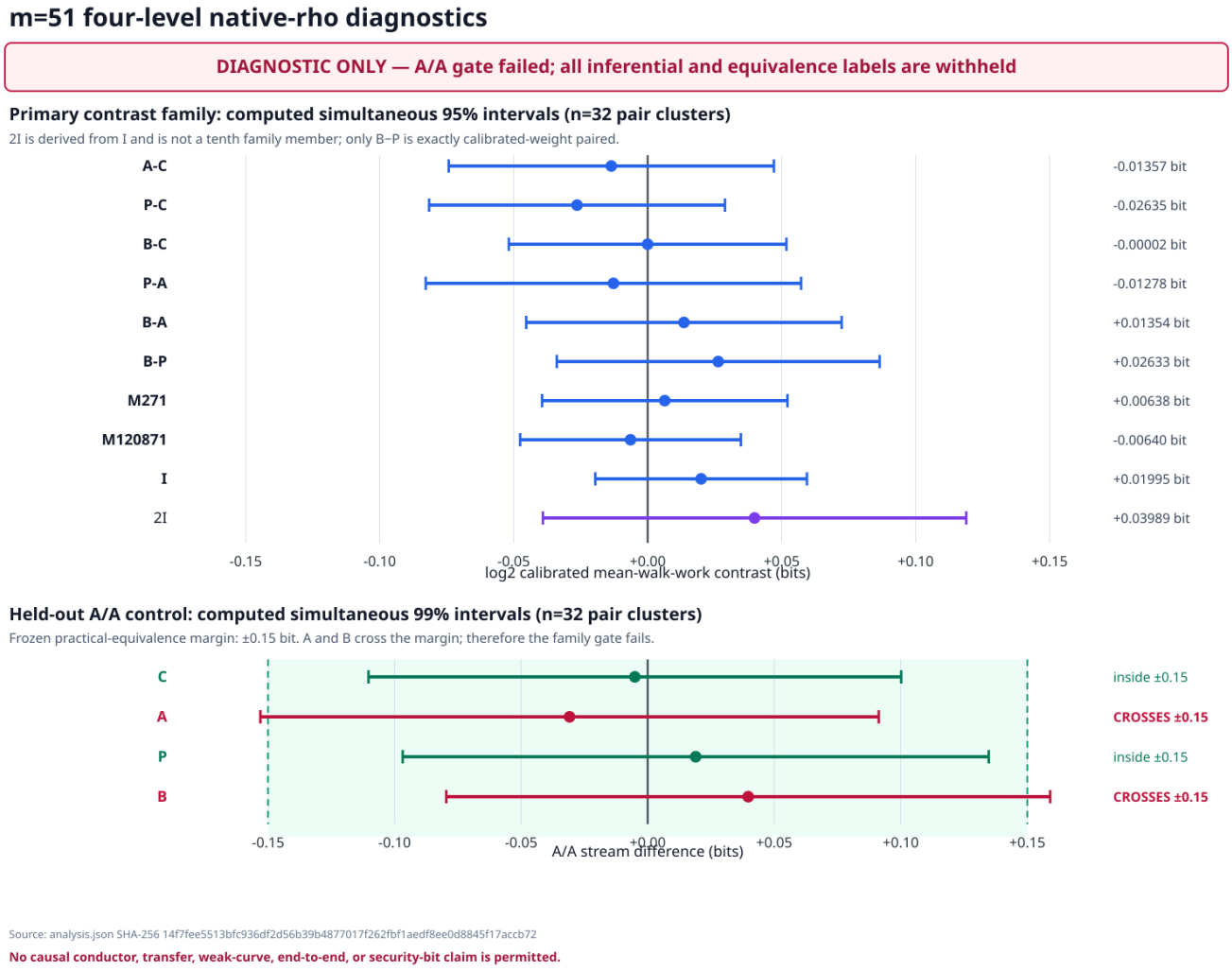
Four-level m=51 native-rho results

Execution and integrity: PASS. Inferential validity: FAIL. The sealed run completed all 20,480 prespecified solves, but the predeclared held-out A/A practical-equivalence gate failed. Therefore **all dependent primary-contrast inferential and equivalence labels are withheld**. The intervals below are reported only as computed diagnostics; they are not admitted confidence/equivalence conclusions.

This is an observations report. It records one bounded native-endpoint experiment and makes no hypothesis transition. It does not establish a weak curve, a causal conductor effect, a transfer advantage, an end-to-end ECDLP cost, a security-bit change, or the absence of any of those things.



Editable vector rendering: four-level-diagnostics.svg, generated by plot_results.py.

Executed scope

The scientific run started at `2026-10-13T13:01:50.540256Z` and finished at `2026-10-13T13:11:58.576041Z` in 608.035806 seconds. It executed 3,556 native setup units over 3,537 distinct models and produced 20,480/20,480 successful records:

- 12,288 primary inference solves: 3,072 for each of `C`, `A`, `P`, and `B`;
- 8,192 held-out A/A solves: 2,048 for each arm;
- 32 selected parent/bottom pair clusters, every one of the 16 `f_E=271` orbits, and all 51 conjugates for every selected `P/B` pair; and
- one native signed-negation action (`mode=0`) throughout, with no point or walk-state transport.

All four arms have ambient order `2,251,799,852,369,764`, prime subgroup order `17,192,986,687`, and cofactor

130,972. The four endomorphism conductors are 1, 271, 120871, and 32756041 = 271*120871.

Measured level summaries

The primary metric is the calibrated arithmetic mean of native online walk_ops after target-dependent table construction. It is not an end-to-end attack cost. Values are for the fixed realized selected-32 endpoint set.

arm	f_E	mean walk_ops	log2(mean)	ratio to C	bits versus C
C	1	122,109.877604	16.897820381	1.000000000	0.000000000
A	271	120,967.009832	16.884254123	0.990640661	-0.013566257
P	120871	119,899.846201	16.871470283	0.981901289	-0.026350098
B	32756041	122,107.843444	16.897796347	0.999983342	-0.000024033

The largest observed pairwise point gap was P-C = -0.026350098 bit, a 0.981901289 work ratio (about 1.81% less measured walk work for P). The large-prime lattice contrast was M120871 = -0.006403937 bit, ratio 0.995570966. The derived 2I diagnostic was +0.039892322 bit; it is twice I, not another family member. The failed A/A gate means this run cannot adjudicate a 3–4-bit hypothesis or promote the observed values into a formal equivalence or absence conclusion.

Exact machine-readable values are in level-summary.csv.

Primary family: labels withheld

The nine-member predeclared family used delete-one-pair jackknife standard errors and a 100,000-draw, per-draw re-studentized Rademacher max-|T| calculation. 2I is twice the family member I, not a tenth hypothesis. Every row below has decision label WITHHELD_AA_ALERT.

contrast	estimate (bit)	work ratio	diagnostic simultaneous 95% interval (bit)
A-C	-0.013566257	0.990640661	[-0.074242757, +0.047110242]
P-C	-0.026350098	0.981901289	[-0.081553057, +0.028852860]
B-C	-0.000024033	0.999983342	[-0.051809003, +0.051760936]
P-A	-0.012783841	0.991178061	[-0.082811235, +0.057243554]
B-A	+0.013542224	1.009430948	[-0.045326134, +0.072410583]
B-P	+0.026326065	1.018415347	[-0.033922530, +0.086574660]
M271	+0.006379904	1.004432005	[-0.039402734, +0.052162541]
M120871	-0.006403937	0.995570966	[-0.047595072, +0.034787199]
I	+0.019946161	1.013921641	[-0.019547072, +0.059439395]
2I, derived	+0.039892322	1.028037095	[-0.039094145, +0.118878789]

Only B-P is exactly calibrated-weight paired. The other rows are balanced common-random-number comparisons. In the fully target/stream-paired common equal-row sensitivity, P-C=-0.022928341, B-A=+0.016260634, M120871=-0.003333854, and I=+0.019594488 bit. This sensitivity is descriptive and is not part of the primary inferential family.

Exact estimates, diagnostic standard errors, adjusted p-values, intervals, scope labels, and pairing flags are in contrast-summary.csv. Per-pair equal-row means and all nine per-pair contrasts are in pair-summary.csv.

Held-out A/A gate

The frozen gate requires every simultaneous 99% A/A interval to lie strictly inside the [-0.15,+0.15] bit practical-equivalence margin. It failed on A and B:

arm	estimate (bit)	diagnostic simultaneous interval (bit)	inside margin?
C	-0.005065263	[-0.110306202, +0.100175675]	yes
A	-0.030866759	[-0.153035934, +0.091302416]	**no**

arm	estimate (bit)	diagnostic simultaneous interval (bit)	inside margin?
P	+0.018975230	[-0.096807690, +0.134758150]	yes
B	+0.039732294	[-0.079573195, +0.159037783]	**no**

A crossed the lower margin by `0.003035934` bit and B crossed the upper margin by `0.009037783` bit. All four intervals include zero, and all four point estimates have absolute value below `0.040` bit. The failed gate cannot distinguish bias from insufficient precision and establishes neither drift nor its absence. The protocol therefore requires the whole primary family to remain unlabeled. A fresh preregistered replication can add evidence, but cannot retroactively validate this run.

Exact gate values are in `aa-summary.csv`, and the preserved decision is `setup_robustness_decision.json`. Because the A/A gate failed, the conditional setup-robustness lane did not trigger. No outcome-driven rerun or backfill occurred.

`REPLICATION_SIZING.md` gives a provisional post-run planning calculation for a separate fresh replication. It is not a frozen protocol and cannot release this run's withheld labels.

Separate setup and process costs

Native setup wall time was recorded separately from `walk_ops`:

arm	invocations	distinct models	mean setup seconds
C	20	1	0.136764462
A	272	272	0.140916414
P	1,632	1,632	0.138110918
B	1,632	1,632	0.138690660

The sum across all 3,556 setup invocations was 492.804730 seconds. Every solve also recorded exactly 162 target-dependent `setup_ops` (128 additions and 34 doublings), excluded from the primary `walk_ops`. Worker elapsed-time maximum was 591.349719 seconds. Wall seconds, group operations, and action primitives have no frozen conversion and are not combined.

Interpretation boundary

The narrow result is: four native endpoint levels were measured successfully, and their calibrated point estimates differed by at most 0.026350098 bit in the pairwise comparisons. The prespecified inference gate failed narrowly, so the result remains inferentially indeterminate.

The experiment supplies no executable isogeny evaluator or dual, no transported subgroup point, and no map construction/evaluation/recovery cost. See `transfer-assessment.json`. It therefore supports neither a source-transfer weakness nor any security-bit statement. One child per sampled parent also means within-parent child heterogeneity is not identified.

The immutable run archive and verification instructions are in `ARCHIVE.md`; independent and post-seal integrity observations are in `AUDIT.md`; unsuccessful or non-triggered paths are retained in `ATTEMPTS.md`. The frozen pre-run `README.md`, `PROTOCOL.md`, and `TRANSFER_BOUNDARY.md` remain unchanged as historical freeze records.